

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1442

International Paper Valliant Ok 345 kV
June 02, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1442. The Interconnection Customer proposes a 494.2 MW natural gas combined cycle facility interconnecting at the International Paper Valliant Ok 345 kV substation in the Virginia area. The request was received on April 02, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 516 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$26,329,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Lamont-Mccain 138.0 kV
Contingency: Artesia Country Club-Clark County 345.0 kV L
Loading: 154.0% (Rating: 205 MVA)
DFAX: 5.1%
Pre-project loading: 141.8%
2. Facility: Mccain-Clark County 138.0 kV
Contingency: Artesia Country Club-Clark County 345.0 kV
Loading: 150.6% (Rating: 219 MVA)
DFAX: 5.1%
Pre-project loading: 139.2%
3. Facility: Lamont-University City 138.0 kV
Contingency: Kearneyville-Clark County 345.0 kV
Loading: 139.5% (Rating: 291 MVA)
DFAX: 5.1%
Pre-project loading: 130.9%
4. Facility: University City-Richord 138.0 kV
Contingency: Clark County-Kearneyville 345.0 kV
Loading: 134.0% (Rating: 330 MVA)
DFAX: 5.1%
Pre-project loading: 126.3%
5. Facility: Sun City-Richford 138.0 kV line
Contingency: Clark County-Kearneyville 345.0 kV
Loading: 132.1% (Rating: 345 MVA)
DFAX: 5.1%
Pre-project loading: 124.7%

6. Facility: Peachland-Sun City 138.0 kV L
Contingency: Clark County-Kearneyville Substation 345.0 kV line
Loading: 131.0% (Rating: 355 MVA)
DFAX: 5.1%
Pre-project loading: 123.9%
7. Facility: Shepherdstown-Peachland 138.0 kV
Contingency: Kearneyville-Clark County 345.0 kV
Loading: 130.5% (Rating: 362 MVA)
DFAX: 5.1%
Pre-project loading: 123.5%
8. Facility: Ross Fork-G-P Gypsum 138.0 kV
Contingency: Artesia Country Club Substation-Air Force Base 345.0 kV
Loading: 119.2% (Rating: 453 MVA)
DFAX: 5.3%
Pre-project loading: 113.4%
9. Facility: Brighton-G-P Gypsum 138.0 kV Line
Contingency: Artesia Country Club-Air Force Base 345.0 kV
Loading: 116.5% (Rating: 505 MVA)
DFAX: 5.7%
Pre-project loading: 110.9%
10. Facility: Artesia Country Club-Ross Fork 138.0 kV Line
Contingency: Air Force Base-Artesia Country Club 345.0 kV
Loading: 113.0% (Rating: 776 MVA)
DFAX: 11.6%
Pre-project loading: 105.6%
11. Facility: Monolith Substation-Santa Rosa Energy Ctr 138.0 kV
Contingency: International Paper Valliant Ok-Kearneyville 345.0 kV
Loading: 104.4% (Rating: 292 MVA)
DFAX: 8.0%
Pre-project loading: 90.9%
12. Facility: N Mesa-Columbus City 138.0 kV line
Contingency: Clark County-Kearneyville 345.0 kV
Loading: 102.8% (Rating: 382 MVA)
DFAX: 11.0%
Pre-project loading: 88.7%
13. Facility: Opequon-Lcec No. 5 138.0 kV
Contingency: Clark County-Kearneyville Substation 345.0 kV
Loading: 101.6% (Rating: 440 MVA)
DFAX: 10.9%
Pre-project loading: 89.4%
14. Facility: Opequon-North Mesa 138.0 kV Line
Contingency: Clark County-Kearneyville Substation 345.0 kV
Loading: 101.2% (Rating: 454 MVA)
DFAX: 10.9%
Pre-project loading: 89.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lamont-Mcain 138.0 kV	Rebuild line with higher-capacity conductor	\$20,322,000	31% (\$6,301,000)
Mccain-Clark County 138.0 kV	Rebuild line with higher-capacity conductor	\$4,495,000	31% (\$1,394,000)
Lamont-University City 138.0 kV	Rebuild line with higher-capacity conductor	\$12,903,000	32% (\$4,104,000)
University City-Richford Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$4,142,000	32% (\$1,318,000)
Richford-Sun City 138.0 kV	Rebuild line with higher-capacity conductor	\$9,441,000	32% (\$3,003,000)
Sun City-Peachland 138.0 kV line	Rebuild line with higher-capacity conductor	\$4,710,000	32% (\$1,498,000)
Shepherdstown-Peachland 138.0 kV	Rebuild line with higher-capacity conductor	\$1,779,000	32% (\$566,000)
Ross Fork-G-P Gypsum 138.0 kV	Rebuild line with higher-capacity conductor	\$5,329,000	42% (\$2,209,000)
Brighton-G-P Gypsum 138.0 kV line	Rebuild line with higher-capacity conductor	\$5,478,000	42% (\$2,309,000)
Artesia Country Club-Ross Fork 138.0 kV Line	Rebuild line with higher-capacity conductor	\$4,499,000	33% (\$1,486,000)
Monolith Substation-Santa Rosa Energy Ctr 138.0 kV	Reconductor line	\$1,798,000	49% (\$887,000)
N Mesa-Columbus City 138.0 kV line	Reconductor line	\$477,000	33% (\$155,000)
Lcec No. 5-Opequon 138.0 kV L	Reconductor line	\$2,168,000	36% (\$779,000)
Opequon-N Mesa 138.0 kV	Reconductor line	\$892,000	36% (\$320,000)
	Total allocated to project		\$26,329,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.